

Chapter 6

Integration Techniques and Improper Integrals (積分技巧與瑕積分)

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Section 6.1

Integration by Parts

(分部、分步或部分積分)



If u and v are diff. functions of x , then

$$(uv)' = u'v + uv' = uv' + vu'.$$

Thus, we immediately obtain

$$\begin{aligned}\int (uv)' dx &= \int uv' dx + \int vu' dx \\ \Rightarrow uv &= \int u dv + \int v du \\ \Rightarrow \int u dv &= uv - \int v du.\end{aligned}$$



Thm 6.1 (I.B.P. 公式)

If $u = u(x)$ and $v = v(x)$ are functions of x having continuous derivatives, then we have

$$\int u \, dv = uv - \int v \, du.$$

How to choose u and dv ?

- $u = u(x)$: easily differentiable function of x . (u : 好微分函數)
- $v = v(x)$: easily integrable function of x . (v : 好積分函數)



Type I of I.B.P.

The integrals of the form

$$\int x^n e^{ax} dx, \quad \int x^n \sin(ax) dx, \quad \int x^n \cos(ax) dx$$

with $n \in \mathbb{R}$ and $a \neq 0$.

How to select u and dv for this case?

- $u = x^n$.
- $dv = e^{ax} dx, \quad \sin(ax) dx, \quad \cos(ax) dx$.



Example 1: Find $\int x e^x dx$

Sol: Let $u = x$ and $dv = e^x dx$

$$\Rightarrow du = dx \text{ and } v = \int e^x dx = e^x.$$

$$\text{So, } \int x e^x dx = \int u dv = uv - \int v du.$$

$$= x e^x - \int e^x dx = \underline{x e^x - e^x + C}$$



Sol: (第二種更簡潔的寫法)

$$\begin{aligned}\int x e^x dx &= \int x d(e^x) \equiv \int u dv \\ &= u v - \int v du \quad (\text{使用 I.B.P. 公式}) \\ &= x e^x - \int e^x dx \\ &= x e^x - e^x + C,\end{aligned}$$

where C is a constant of integration.



Example 4: Find $\int x^2 \sin x dx$ (使用兩次 I.B.P.)

Sol: Let $u = x^2$ and $dv = \sin x dx$.

$$\Rightarrow du = 2x dx \text{ and } v = \int \sin x dx = -\cos x.$$

$$\begin{aligned} \Rightarrow \int x^2 \sin x dx &= \int u dv = uv - \int v du = -x^2 \cos x - \int 2x(-\cos x) dx \\ &= -x^2 \cos x + 2 \int x \cos x dx \quad (*) \end{aligned}$$

Let $u = x$ and $dv = \cos x dx$

$$\Rightarrow du = dx \text{ and } v = \int \cos x dx = \sin x.$$

$$\begin{aligned} \text{From } (*) \Rightarrow \int x^2 \sin x dx &= -x^2 \cos x + 2 \int u dv \\ &= -x^2 \cos x + 2 \left(x \sin x - \int \sin x dx \right) \text{ by I.B.P.} \\ &= -x^2 \cos x + 2 x \sin x + 2 \cos x + C \end{aligned}$$



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Example 7 (Tabular Method 表格法)Find $\int x^2 \sin 4x dx$ using the tabular method.Sol:

+	x^2	$\sin 4x dx$
-	$2x$	$\frac{1}{4} \cos 4x$
+	2	$-\frac{1}{16} \sin 4x$
-	0	$\frac{1}{64} \cos 4x$

$$\text{So, } \int x^2 \sin 4x dx = -\frac{x^2}{4} \cos 4x + \frac{x}{8} \sin 4x + \frac{1}{32} \cos 4x + C$$



Type II of I.B.P.

The integrals of the form

$$\int x^n \ln x \, dx, \quad \int x^n \sin^{-1} x \, dx, \quad \int x^n \tan^{-1} x \, dx$$

with $n \neq -1$.

How to select u and dv for this case?

- $u = \ln x, \quad \sin^{-1} x, \quad \tan^{-1} x.$
- $dv = x^n dx.$



Example 2: Find $\int x^2 \ln x \, dx$.

Sol: Let $u = \ln x$ and $dv = x^2 \, dx$
 $\Rightarrow du = \frac{1}{x} \, dx$ and $v = \frac{1}{3} x^3$.

$$\text{So, } \int x^2 \ln x \, dx = \int u \, dv = uv - \int v \, du.$$

$$= \frac{1}{3} x^3 \ln x - \int \frac{1}{3} x^3 \left(\frac{1}{x} \right) dx = \frac{1}{3} x^3 \ln x - \frac{1}{3} \int x^2 \, dx.$$

$$= \frac{x^3}{3} \ln x - \frac{1}{9} x^3 + C$$



Example 3: Find $\int_0^1 \sin^{-1} x \, dx$.

Sol: Let $u = \sin^{-1} x$ and $dv = dx$.

$$\Rightarrow du = \frac{1}{\sqrt{1-x^2}} dx \quad \text{and} \quad v = x.$$

$$\begin{aligned} \Rightarrow \int \sin^{-1} x \, dx &= x \sin^{-1} x - \int \frac{x}{\sqrt{1-x^2}} \, dx. \\ &= x \sin^{-1} x + \frac{1}{2} \int \frac{-2x}{\sqrt{1-x^2}} \, dx \\ &= x \sin^{-1} x + \sqrt{1-x^2} + C \end{aligned}$$

$$\text{So, } \int_0^1 \sin^{-1} x \, dx = \left(x \sin^{-1} x + \sqrt{1-x^2} \right) \Big|_0^1 = \frac{\pi}{2} - 1.$$



88 Example 5 Find $\int \sec^3 x dx$ (重要公式, 必記!!)

Sol: $\int \sec^3 x dx = \int \sec \cdot \underline{\sec^2 x} dx = \int \sec x \left(\frac{1}{\cos^2 x} \right) d(\tan x)$

$$= \sec x \tan x - \int \tan x d(\sec x)$$

$$= \sec x \tan x - \int \sec x \tan^2 x dx.$$

$$= \sec x \tan x - \int \sec x (\sec^2 - 1) dx. \quad \left(\because \tan^2 x + 1 = \sec^2 x \right)$$

$$= \sec x \tan x - \underbrace{\int \sec^3 x dx} + \int \sec x dx.$$

$$\text{So, } 2 \int \sec^3 x dx = \sec x \tan x + \int \sec x dx$$

$$\Rightarrow \underline{\int \sec^3 x dx = \frac{1}{2} \sec x \tan x + \frac{1}{2} \ln |\sec x + \tan x| + C}$$



Type III of I.B.P.

The integrals of the form

$$\int e^{ax} \sin(bx) dx, \quad \int e^{ax} \cos(bx) dx$$

with $a \neq 0$ and $b \neq 0$.

How to select u and dv for this case?

- $u = e^{ax}$ and $dv = \sin(bx)dx, \quad \cos(bx)dx$.
- $u = \sin(bx), \quad \cos(bx)$ and $dv = e^{ax}dx$.
- 通常使用兩次 I.B.P. 公式!



Example: Find $\int e^x \cos x dx$

Sol: $\int e^x \cos x dx = \int e^x d(\sin x)$

$$= e^x \sin x - \int \sin x d(e^x) = e^x \sin x - \int e^x \sin x dx$$

$$= e^x \sin x + \int e^x d(\cos x) = e^x \sin x + e^x \cos x - \int e^x \cos x dx$$

$$\text{So, } 2 \int e^x \cos x dx = e^x (\sin x + \cos x)$$

$$\Rightarrow \int e^x \cos x dx = \frac{e^x}{2} (\sin x + \cos x) + C$$



Section 6.2

Trigonometric Integrals

(三角積分)



Main Goals

Try to find the integrals of the form

(INT-1) : $\int \sin^m x \cos^n x dx$, where $m, n \in \mathbb{Q}$.

(INT-2) : $\int \sec^m x \tan^n x dx$, where $m, n \in \mathbb{Q}$.

(INT-3) : $\int \sin \alpha \cos \beta dx$, $\int \sin \alpha \sin \beta dx$, $\int \cos \alpha \cos \beta dx$,
where $\alpha = \alpha(x)$ and $\beta = \beta(x)$ are functions of x .



Type I of (INT-1)

If $m = 2k + 1$ is odd for some $k \in \mathbb{N}$, then

$$\begin{aligned}\int \sin^{2k+1} x \cos^n x dx &= \int (\sin^2 x)^k \cos^n x \cdot \sin x dx \\ &= - \int (1 - \cos^2 x)^k \cos^n x d(\cos x) = - \int (1 - u^2)^k u^n du,\end{aligned}$$

where we let $u = \cos x$.



Example 1 (Type I a (3) 子) Find $\int \sin^3 x \cos^4 x dx$.

Sol: $\int \sin^3 x \cos^4 x dx = \int \sin^2 x \cos^4 x \sin x dx$

$$= - \int (1 - \cos^2 x) \cos^4 x d(\cos x) = \int (\cos^6 x - \cos^4 x) d(\cos x).$$

$$= \frac{\cos^7 x}{7} - \frac{\cos^5 x}{5} + C.$$

Per-Dust



Type II of (INT-1)

If $n = 2k + 1$ is odd for some $k \in \mathbb{N}$, then

$$\begin{aligned}\int \sin^m x \cos^{2k+1} x \, dx &= \int \sin^m x (\cos^2 x)^k \cdot \cos x \, dx \\ &= \int \sin^m x (1 - \sin^2 x)^k d(\sin x) = \int u^m (1 - u^2)^k du,\end{aligned}$$

where we let $u = \sin x$.



Example 2: Find $\int_{\pi/6}^{\pi/3} \frac{\cos^3 x}{\sqrt{\sin x}} dx = \int_{\pi/6}^{\pi/3} (\sin x)^{-1/2} \cos^3 x dx.$

Sol. $\int_{\pi/6}^{\pi/3} (\sin x)^{-1/2} \cos^2 x \cdot \cos x dx = \int_{\pi/6}^{\pi/3} (\sin x)^{-1/2} (1 - \sin^2 x) d(\sin x)$

$= \int_{\pi/6}^{\pi/3} \left[(\sin x)^{-1/2} - (\sin x)^{3/2} \right] d(\sin x).$

$= \left[2(\sin x)^{1/2} - \frac{2}{5}(\sin x)^{5/2} \right]_{\pi/6}^{\pi/3}$

$= 2\left(\frac{\sqrt{3}}{2}\right)^{1/2} - \frac{2}{5}\left(\frac{\sqrt{3}}{2}\right)^{5/2} - \sqrt{2} \cancel{\frac{\sqrt{2}}{20}} + \frac{\sqrt{2}}{20}$



Type III of (INT-1)

If m and n are even and nonnegative, try to use the identities

$$\sin^2 x = \frac{1 - \cos(2x)}{2}, \quad \cos^2 x = \frac{1 + \cos(2x)}{2}.$$

(當 m, n 為偶數或零，試用倍角公式!)



Example 3: Find $\int \cos^4 x dx$ with $m=0$ and $n=4$.

Sol: $\int \cos^4 x dx = \int \left(\frac{1 + \cos 2x}{2} \right)^2 dx = \frac{1}{4} \int (1 + 2\cos 2x + \cos^2 2x) dx$

$= \frac{1}{4} \int \left(1 + 2\cos 2x + \frac{1 + \cos 4x}{2} \right) dx = \frac{1}{4} \int \left(\frac{3}{2} + 2\cos 2x + \frac{1}{2} \cos 4x \right) dx.$

$= \int \left(\frac{3}{8} + \frac{1}{2} \cos 2x + \frac{1}{8} \cos 4x \right) dx$

$= \frac{3x}{8} + \frac{1}{4} \sin 2x + \frac{1}{32} \sin 4x + C.$



Type I of (INT-2)

If $m = 2k$ is even for some $k \in \mathbb{N}$, then

$$\begin{aligned}\int \sec^{2k} x \tan^n x dx &= \int (\sec^2 x)^{k-1} \tan^n x \cdot \sec^2 x dx \\ &= \int (1 + \tan^2 x)^{k-1} \tan^n x d(\tan x) = \int (1 + u^2)^{k-1} u^n du,\end{aligned}$$

where we let $u = \tan x$.



Example 5: Find $\int \sec^4(3x) \tan^3(3x) dx$.

Sol: ~~$\int \sec^2(3x) \tan(3x) \sec^2(3x) dx$~~ Let $u = \tan(3x)$. Then

$$du = 3 \sec^2(3x) dx \Rightarrow \sec^2(3x) dx = \frac{1}{3} du$$

$$\text{So, } \int \sec^4(3x) \tan^3(3x) dx = \int \sec^2(3x) \tan^3(3x) \cdot \sec^2(3x) dx$$

$$= \int [1 + \tan^2(3x)] \tan^3(3x) \cdot \sec^2(3x) dx = \int (1 + u^2) u^3 du.$$

$$= \frac{1}{3} \int (u^3 + u^5) du = \frac{\tan^4(3x)}{12} + \frac{\tan^6(3x)}{18} + C$$



Type II of (INT-2)

If $n = 2k + 1$ is odd for some $k \in \mathbb{N}$, then

$$\begin{aligned}\int \sec^m x \tan^{2k+1} x \, dx &= \int \sec^{m-1} x (\tan^2)^k \cdot \sec x \tan x \, dx \\ &= \int \sec^{m-1} x (\sec^2 - 1)^k d(\sec x) = \int u^{m-1} (u^2 - 1)^k du,\end{aligned}$$

where we let $u = \sec x$.



Example 4: Find $\int \frac{\tan^3 x}{\sec x} dx = \int (\sec x)^{-1/2} \tan^3 x dx.$

Sol: $\int (\sec x)^{-1/2} \tan^3 x dx = \int (\sec x)^{-3/2} \tan^2 x (\sec x \tan x) dx$

$$= \int (\sec x)^{-3/2} (\sec^2 x - 1) d(\sec x) = \int \left[(\sec x)^{1/2} - (\sec x)^{-3/2} \right] d(\sec x)$$

$$= \frac{2}{3} (\sec x)^{3/2} + 2 (\sec x)^{-1/2} + C$$



Type III of (INT-2)

If m is odd or n is even, try to use the identity

$$\tan^2 x = \sec^2 x - 1.$$

(當 m 是奇數或 n 是偶數時，試用上述等式!)



Example 6: Find $\int_0^{\pi/4} \tan^4 x \, dx$.

Sol: $\int_0^{\pi/4} \tan^2 x \cdot \tan^2 x \, dx = \int_0^{\pi/4} (\sec^2 x - 1) \tan^2 x \, dx$

$$= \int_0^{\pi/4} \sec^2 x \tan^2 x \, dx - \int_0^{\pi/4} (\sec^2 x - 1) \, dx.$$

$$= \left(\frac{\tan^3 x}{3} - \tan x + x \right) \Big|_0^{\pi/4} = \frac{1}{3} + \frac{\pi}{4} \approx 0.119$$



Integrals of Sine-Cosine Products

if $\alpha = \alpha(x)$ and $\beta = \beta(x)$ are conti. functions of x , how to evaluate

(INT-3) : $\int \sin \alpha \cos \beta \, dx, \quad \int \sin \alpha \sin \beta \, dx, \quad \int \cos \alpha \cos \beta \, dx$

using the product-to-sum identities (積化和差等式)?



Product-to-Sum Identities

$$(1) \sin \alpha \sin \beta = \frac{1}{2} [\cos(\alpha - \beta) - \cos(\alpha + \beta)].$$

$$(2) \cos \alpha \cos \beta = \frac{1}{2} [\cos(\alpha - \beta) + \cos(\alpha + \beta)].$$

$$(3) \sin \alpha \cos \beta = \frac{1}{2} [\sin(\alpha - \beta) + \sin(\alpha + \beta)].$$



Example 8*: (積化和差の公式) 202

Find $\int \sin(5x) \cos(4x) dx$. with $\alpha=5x$ and $\beta=4x$.

Sol: $\int \sin(5x) \cos(4x) dx = \frac{1}{2} \int [\sin(5x-4x) + \sin(5x+4x)] dx$

$$= \frac{1}{2} \int (\sin x + \sin 9x) dx = \frac{-\cos x}{2} - \frac{\cos(9x)}{18} + C$$



Section 6.3

Trigonometric Substitution

(三角代換)



Main Goal

- To deal with the integrals involving

$$\sqrt{a^2 - x^2}, \quad \sqrt{a^2 + x^2} \quad \text{or} \quad \sqrt{x^2 - a^2}$$

with $a > 0$.

- More precisely, how to evaluate the three types of aforementioned integrals using **the technique of trigonometric substitution?**



Type I of Trigonometric Substitution

For integrals involving $\sqrt{a^2 - x^2}$ with $a > 0$, let

$$x = a \sin \theta.$$

Then $dx = a \cos \theta d\theta$ and we see that

$$\sqrt{a^2 - x^2} = \sqrt{a^2(1 - \sin^2 \theta)} = \sqrt{a^2 \cos^2 \theta} = a \cos \theta$$

for $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$.



Example 1: Find $\int \frac{1}{x^2 \sqrt{9-x^2}} dx$.

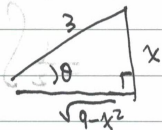
Sol: Let $x = 3 \sin \theta$ for $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$. Then

$$dx = 3 \cos \theta d\theta \text{ and } \sqrt{9-x^2} = 3 \cos \theta.$$

$$\text{So, } \int \frac{dx}{x^2 \sqrt{9-x^2}} = \int \frac{\cancel{3 \cos \theta}}{(9 \sin^2 \theta)(\cancel{3 \cos \theta})} d\theta.$$

$$= \frac{1}{9} \int \csc^2 \theta d\theta = -\frac{1}{9} \cot \theta + C$$

$$= \frac{1}{9} \left(\frac{\sqrt{9-x^2}}{x} \right) + C = \frac{\sqrt{9-x^2}}{9x} + C$$



Type II of Trigonometric Substitution

For integrals involving $\sqrt{a^2 + x^2}$ with $a > 0$, let

$$x = a \tan \theta.$$

Then $dx = a \sec^2 \theta d\theta$ and we see that

$$\sqrt{a^2 + x^2} = \sqrt{a^2(1 + \tan^2 \theta)} = \sqrt{a^2 \sec^2 \theta} = a \sec \theta$$

for $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$.



Example 2: Find $\int \frac{dx}{\sqrt{4x^2+1}}$.



Sol: Note that $\int \frac{dx}{\sqrt{4x^2+1}} = \frac{1}{2} \int \frac{dx}{\sqrt{x^2+\frac{1}{4}}}$

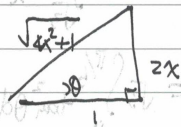
Let $x = \frac{1}{2} \tan \theta$. Then $dx = \frac{1}{2} \sec^2 \theta d\theta$ and

$\sqrt{x^2 + \frac{1}{4}} = \sqrt{\frac{1}{4} \sec^2 \theta} = \frac{1}{2} \sec \theta$ for $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$.

So, $\int \frac{\frac{1}{2} dx}{\sqrt{x^2 + \frac{1}{4}}} = \int \frac{(\frac{1}{2})(\frac{1}{2}) \sec^2 \theta}{(\frac{1}{2}) \sec \theta} d\theta$

$= \frac{1}{2} \int \sec \theta d\theta = \frac{1}{2} \ln |\sec \theta + \tan \theta| + C$

$= \frac{1}{2} \ln |\sqrt{4x^2+1} + 2x| + C$



Example 3: Find $\int \frac{dx}{\sqrt{4x^2+1}}$ ($\tan \theta = 2x$)

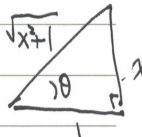


Example 3: Find $\int \frac{dx}{(x^2+1)^{3/2}}$ ($\tan\theta = x$).

Sol: Let $x = \tan\theta$ for $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$. Then

$$\int \frac{dx}{(x^2+1)^{3/2}} = \int \frac{\sec^2\theta d\theta}{(\tan^2\theta+1)^{3/2}} = \int \frac{\sec^2\theta}{\sec^3\theta} d\theta$$

$$= \int \frac{d\theta}{\sec\theta} = \int \cos\theta d\theta = \sin\theta + C = \frac{x}{\sqrt{x^2+1}} + C.$$



($\tan\theta = x$)



Type III of Trigonometric Substitution

For integrals involving $\sqrt{x^2 - a^2}$ with $a > 0$, let

$$x = a \sec \theta.$$

Then $dx = a \sec \theta \tan \theta d\theta$ and we see that

$$\sqrt{x^2 - a^2} = \sqrt{a^2(\sec^2 \theta - 1)} = \sqrt{a^2 \tan^2 \theta} = a |\tan \theta|$$

for $0 \leq \theta < \frac{\pi}{2}$ or $\frac{\pi}{2} < \theta \leq \pi$.



Example 4: Find $\int \frac{2\sqrt{x^2-3}}{x} dx$.

Sol: Let $x = \sqrt{3} \sec \theta$ for $\theta \in [0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$.

Then $dx = \sqrt{3} \sec \theta \tan \theta d\theta$, $\sqrt{x^2-3} = \sqrt{3 \sec^2 \theta - 3} = \sqrt{3 \tan^2 \theta} = \sqrt{3} \tan \theta$

θ	0	$\frac{\pi}{6}$
x	$\sqrt{3}$	2

and $\sqrt{x^2-3} = \sqrt{3(\sec^2 \theta - 1)} = \sqrt{3 \tan^2 \theta} = \sqrt{3} \tan \theta$ for $0 \leq \theta \leq \frac{\pi}{6}$.

Hence, $\int \frac{2\sqrt{x^2-3}}{x} dx = \int_0^{\pi/6} \frac{(\sqrt{3} \tan \theta)(\sqrt{3} \sec \theta \tan \theta)}{\sqrt{3} \sec \theta} d\theta$.

$= \sqrt{3} \int_0^{\pi/6} \tan^2 \theta d\theta = \sqrt{3} \int_0^{\pi/6} (\sec^2 \theta - 1) d\theta$.

$= \sqrt{3} (\tan \theta - \theta) \Big|_0^{\pi/6} = \sqrt{3} \left(\frac{1}{\sqrt{3}} - \frac{\pi}{6} \right) = 1 - \frac{\sqrt{3}\pi}{6}$.



Section 6.4

Partial Fractions

(部分分式)



Long Division (長除法)

If $N(x)$ and $D(x)$ are polynomials with $\deg(N) \geq \deg(D)$, then

$$\frac{N(x)}{D(x)} = Q(x) + \frac{N_1(x)}{D(x)},$$

where $Q(x)$ and $N_1(x)$ are polynomials with $\deg(N_1) < \deg(D)$. In this case, $\frac{N_1(x)}{D(x)}$ is called a **proper** rational function of x (真分式).

Note: integrating w.r.t. $x \implies$

$$\int \frac{N(x)}{D(x)} dx = \int Q(x) dx + \int \frac{N_1(x)}{D(x)} dx.$$



Remarks

- $\int Q(x) dx$ can be evaluated easily by the **power rule**.
- But, how to evaluate the integral $\int \frac{N_1(x)}{D(x)} dx$ using the technique of **partial fraction decomposition**?



Partial Fraction Decomposition

If $D(x) = (px + q)^m(ax^2 + bx + c)^n$ with $p \neq 0$ and $b^2 - 4ac < 0$, may write

$$\frac{N_1(x)}{D(x)} = \frac{A_1}{px + q} + \frac{A_2}{(px + q)^2} + \cdots + \frac{A_m}{(px + q)^m} \\ + \frac{B_1x + C_1}{ax^2 + bx + c} + \frac{B_2x + C_2}{(ax^2 + bx + c)^2} + \cdots + \frac{B_nx + C_n}{(ax^2 + bx + c)^n},$$

where A_i ($i = 1, \dots, m$) and B_j, C_j ($j = 1, \dots, n$) are unknowns.



Type I: Distinct or Repeated Linear Factors

The denominator (分母) $D(x)$ contains

- distinct linear factors (相異的一次因式)

$$(p_1x + q_1)(p_2x + q_2) \cdots (p_mx + q_m)$$

- repeated linear factors (重複的一次因式)

$$(px + q)^m,$$

where $p_j \neq 0$ ($j = 1, 2, \dots, m$) and $p \neq 0$ for some $m \in \mathbb{N}$.



Example 1: (相異の一次因式)

Write the partial fraction decomposition for $\frac{1}{x^2-5x+6}$.

Sol: $\frac{1}{x^2-5x+6} = \frac{1}{(x-3)(x-2)} = \frac{A}{x-3} + \frac{B}{x-2}$

$$\Rightarrow 1 = A(x-2) + B(x-3) = (A+B)x + (-2A-3B)$$

$$\Rightarrow \begin{cases} A+B=0 \\ -2A-3B=1 \end{cases} \Rightarrow B=-1 \text{ and } A=1$$

Sol, $\frac{1}{x^2-5x+6} = \frac{1}{x-3} - \frac{1}{x-2}$ *



Example 2: (重複 α -次因式)

Find $\int \frac{5x^3 + 20x + 6}{x^3 + 2x^2 + x} dx$.

Sol: Let $f(x) = \frac{5x^3 + 20x + 6}{x^3 + 2x^2 + x}$. Since $x^3 + 2x^2 + x = x(x+1)^2$,

may write $f(x) = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$.

$$\Rightarrow 5x^2 + 20x + 6 = A(x+1)^2 + Bx(x+1) + Cx$$

$$= (A+B)x^2 + (2A+B+C)x + A$$



$$\Rightarrow A=6, 2A+B+C=20 \text{ and } A+B=5$$

$$\Rightarrow A=6, B=-1, C=9. \Rightarrow f(x) = \frac{6}{x} - \frac{1}{x+1} + \frac{9}{(x+1)^2}$$

$$\begin{aligned} \text{So, } \int f(x) dx &= \int \left(\frac{6}{x} - \frac{1}{x+1} + \frac{9}{(x+1)^2} \right) dx = 6 \ln|x| - \ln|x+1| - \frac{9}{x+1} + C \\ &= \ln \left| \frac{x^6}{x+1} \right| - \frac{9}{x+1} + C. \end{aligned}$$

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Type II: Distinct or Repeated Quadratic Factors (二次因式)

The denominator $D(x)$ contains

- distinct quadratic factors

$$(a_1x^2 + b_1x + c_1)(a_2x^2 + b_2x + c_2) \cdots (a_nx^2 + b_nx + c_n)$$

- repeated quadratic factors

$$(ax^2 + bx + c)^n,$$

where $b_j^2 - 4a_jc_j < 0$ ($j = 1, 2, \dots, n$) and $b^2 - 4ac < 0$ for some $n \in \mathbb{N}$.

Note: these quadratic factors are called **irreducible** (不可既約)!



Example 3: (相異なる一次と二次因式)

Find $\int \frac{2x^3 - 4x - 8}{(x^2 - x)(x^2 + 4)} dx$.

Sol: Let $f(x) = \frac{2x^3 - 4x - 8}{(x^2 - x)(x^2 + 4)}$. Since $(x^2 - x)(x^2 + 4) = x(x-1)(x^2 + 4)$,

write $f(x) = \frac{A}{x} + \frac{B}{x-1} + \frac{Cx+D}{x^2+4}$.



$$\Rightarrow 2x^3 - 4x - 8 = A(x-1)(x^2+4) + Bx(x^2+4) + (Cx+D)(x^2-x)$$

Take $x=0$ in (*) $\Rightarrow -8 = A(-1)(4) \Rightarrow \underline{A=2}$

Take $x=1$ in (*) $\Rightarrow 2-4-8 = 5B \Rightarrow \underline{B=-2}$

So, $2x^3 - 4x - 8 = 2(x-1)(x^2+4) + (-2)x(x^2+4) + (Cx+D)(x^2-x)$

Take $x=-1$ $\Rightarrow 2 = -C+D$

Take $x=2$ $\Rightarrow 8 = 2C+D$

$\} \Rightarrow \underline{C=2}$ and $\underline{D=4}$

So, $\int f(x) dx = \int \left(\frac{2}{x} - \frac{2}{x-1} + \frac{2x+4}{x^2+4} \right) dx$

$$= \int \frac{2}{x} dx - \int \frac{2}{x-1} dx + \int \frac{2x}{x^2+4} dx + \int \frac{4}{x^2+4} dx$$

$$= 2 \ln|x| - 2 \ln|x-1| + \ln(x^2+4) + 2 \tan^{-1}\left(\frac{x}{2}\right) + C$$



Example 4: (重複の二次因式)

Find $\int \frac{8x^3 + 13x}{(x^2 + 2)^2} dx$.

Sol: Write $f(x) = \frac{8x^3 + 13x}{(x^2 + 2)^2} = \frac{Ax + B}{x^2 + 2} + \frac{Cx + D}{(x^2 + 2)^2}$.

$$(Ax + B)(x^2 + 2) + (Cx + D) = 8x^3 + 13x$$

$$= Ax^3 + Bx^2 + (2A + C)x + (2B + D).$$



$$\Rightarrow A=8, B=0, 2A+C=13 \text{ and } 2B+D=0.$$

$$\Rightarrow A=8, B=0, C=-3, D=0.$$

$$\text{So, } \int f(x) dx = \int \left(\frac{8x}{x^2+2} + \frac{-3x}{(x^2+2)^2} \right) dx.$$

$$= 4 \int \frac{2x}{x^2+2} dx - \frac{3}{2} \int \frac{2x}{(x^2+2)^2} dx.$$

$$= 4 \ln(x^2+2) + \frac{3}{2(x^2+2)} + C.$$



Section 6.7

Improper Integrals

(瑕積分)



Two Types of Improper Integrals

Type I : Infinite Limits of Integration (無窮積分上下限), e.g.,

$$\int_a^{\infty} f(x) dx \quad \text{or} \quad \int_{-\infty}^b f(x) dx.$$

Type II : Infinite Discontinuities (無窮不連續點), e.g., we say that

$\int_a^b f(x) dx$ is an improper integral of Type II if

$$\lim_{x \rightarrow c^+} f(x) = \pm\infty \quad \text{or} \quad \lim_{x \rightarrow c^-} f(x) = \pm\infty$$

for some $c \in [a, b]$.



Type I: Infinite Limits of Integration (1/2)

- (1) If f is conti. on $[a, \infty)$, the improper integral

$$\int_a^{\infty} f(x) dx = \lim_{b \rightarrow \infty} \left[\int_a^b f(x) dx \right]$$

converges (收斂) whenever the limit exists. Otherwise, we say that the improper integral diverges (發散).

- (2) If f is conti. on $(-\infty, b]$, the improper integral

$$\int_{-\infty}^b f(x) dx = \lim_{a \rightarrow -\infty} \left[\int_a^b f(x) dx \right]$$

converges (收斂) whenever the limit exists. Otherwise, we say that the improper integral diverges (發散).



Type I: Infinite Limits of Integration (2/2)

(3) If f is conti. on $(-\infty, \infty)$, the improper integral

$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^c f(x) dx + \int_c^{\infty} f(x) dx$$

converges (收斂) whenever the improper integrals on the RHS both converge for all $c \in \mathbb{R}$.

Note: we say that the improper integral $\int_{-\infty}^{\infty} f(x) dx$ diverges if either of the improper integrals on the RHS diverges.



Example 2: Evaluate the improper integral.

$$\begin{aligned} \text{(a)} \quad \int_0^{\infty} e^{-x} dx &= \lim_{b \rightarrow \infty} \left[\int_0^b e^{-x} dx \right] = \lim_{b \rightarrow \infty} \left(-e^{-x} \Big|_0^b \right) \\ &= \lim_{b \rightarrow \infty} (-e^{-b} + 1) = \underline{1}. \quad \# \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \int_0^{\infty} \frac{1}{x^2+1} dx &= \lim_{b \rightarrow \infty} \left(\int_0^b \frac{dx}{x^2+1} \right) = \lim_{b \rightarrow \infty} \left(\tan^{-1} x \Big|_0^b \right) \\ &= \lim_{b \rightarrow \infty} (\tan^{-1} b - 0) = \underline{\frac{\pi}{2}}. \quad \# \end{aligned}$$



Example 4: Evaluate $\int_{-\infty}^{\infty} \frac{e^x}{1+e^{2x}} dx$.

Sol: $\int \frac{e^x}{1+e^{2x}} dx = \int \frac{1}{1+(e^x)^2} d(e^x) = \tan^{-1}(e^x) + C$.

$$\int_{-\infty}^{\infty} \frac{e^x}{1+e^{2x}} dx = \int_{-\infty}^0 \frac{e^x}{1+e^{2x}} dx + \int_0^{\infty} \frac{e^x}{1+e^{2x}} dx.$$

$$= \lim_{a \rightarrow -\infty} \left[\tan^{-1}(e^x) \Big|_a^0 \right] + \lim_{b \rightarrow \infty} \left[\tan^{-1}(e^x) \Big|_0^b \right].$$

$$= \lim_{a \rightarrow -\infty} \left[\frac{\pi}{4} - \tan^{-1}(e^a) \right] + \lim_{b \rightarrow \infty} \left[\tan^{-1}(e^b) - \frac{\pi}{4} \right]$$

$$= \left(\frac{\pi}{4} - 0 \right) + \left(\frac{\pi}{2} - \frac{\pi}{4} \right) = \underline{\underline{\frac{\pi}{2}}}. \quad \times$$



Type II: Infinite Discontinuities (1/2)

- (1) If f is conti. on $[a, b)$ and $\lim_{x \rightarrow b^-} f(x) = \pm\infty$, the integral

$$\int_a^b f(x) dx = \lim_{c \rightarrow b^-} \left[\int_a^c f(x) dx \right]$$

converges (收斂) whenever the limit exists. Otherwise, we say that the improper integral diverges (發散).

- (2) If f is conti. on $(a, b]$ and $\lim_{x \rightarrow a^+} f(x) = \pm\infty$, the integral

$$\int_a^b f(x) dx = \lim_{c \rightarrow a^+} \left[\int_c^b f(x) dx \right]$$

converges (收斂) whenever the limit exists. Otherwise, we say that the improper integral diverges (發散).



Type II: Infinite Discontinuities (2/2)

- (3) If f is conti. on $[a, b]$ and has an infinite discontinuity at $c \in (a, b)$, the improper integral

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

converges (收斂) whenever the improper integrals on the RHS both converge.

Note: we say that the improper integral $\int_a^b f(x) dx$ diverges if either of the improper integrals on the RHS diverges.



Example 5: Evaluate $\int_0^1 \frac{1}{\sqrt[3]{x}} dx = \int_0^1 x^{-1/3} dx$. *

Sol.: $\because \lim_{x \rightarrow 0^+} \frac{1}{\sqrt[3]{x}} = \left(\frac{1}{0} \right) = \infty$ and $f(x) = \frac{1}{\sqrt[3]{x}}$ is cont. on $(0, 1]$.

$\therefore \int_0^1 x^{-1/3} dx$ is an improper integral of Type II.

$$\text{So, } \int_0^1 x^{-1/3} dx = \lim_{a \rightarrow 0^+} \left(\int_a^1 x^{-1/3} dx \right) = \lim_{a \rightarrow 0^+} \left(\frac{3}{2} x^{2/3} \Big|_a^1 \right) = \frac{3}{2}.$$



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Example 7: Evaluate $\int_{-1}^2 \frac{1}{x^3} dx$.

Sol: Note that $\int_{-1}^2 x^{-3} dx = \frac{-1}{2} x^{-2} \Big|_{-1}^2 = \dots (X)$



$$\lim_{x \rightarrow 0^+} \frac{1}{x^3} = \infty \text{ and } \lim_{x \rightarrow 0^-} \frac{1}{x^3} = -\infty.$$

$\int_{-1}^2 x^{-3} dx$ is an improper integral of Type II.

$$\Rightarrow \int_{-1}^2 x^{-3} dx = \int_{-1}^0 x^{-3} dx + \int_0^2 x^{-3} dx.$$

$$= \lim_{b \rightarrow 0^-} \left(\frac{-1}{2} x^{-2} \Big|_{-1}^b \right) + \lim_{a \rightarrow 0^+} \left(\frac{-1}{2} x^{-2} \Big|_a^2 \right)$$

$$= \lim_{b \rightarrow 0^-} \left(\frac{-1}{2b^2} + \frac{1}{2} \right) + \lim_{a \rightarrow 0^+} \left(\frac{-1}{8} + \frac{1}{2a^2} \right) \text{ diverges,}$$

$$\text{Since } \int_{-1}^0 \frac{1}{x^3} dx = \lim_{b \rightarrow 0^-} \left(\frac{-1}{2b^2} + \frac{1}{2} \right) = -\infty \text{ and } \int_0^2 \frac{1}{x^3} dx = \infty. \#$$



Example 8 (Type I & Type II 的積分技巧同時出現)

Evaluate $\int_0^{\infty} \frac{dx}{\sqrt{x}(x+1)}$

Sol: Let $u = \sqrt{x}$ for $x \geq 0$. Then $du = \frac{1}{2\sqrt{x}} dx$ or $\frac{dx}{\sqrt{x}} = 2du$.

$$\Rightarrow \int \frac{dx}{\sqrt{x}(x+1)} = \int \frac{2du}{u^2+1} = 2 \tan^{-1} u + C = 2 \tan^{-1} \sqrt{x} + C \quad (*)$$



$$\circ \circ \lim_{x \rightarrow 0^+} \frac{1}{\sqrt{x}(x+1)} = \infty$$

$$\circ \int_0^{\infty} \frac{dx}{\sqrt{x}(x+1)} = \int_0^1 \frac{dx}{\sqrt{x}(x+1)} + \int_1^{\infty} \frac{dx}{\sqrt{x}(x+1)} = \lim_{a \rightarrow 0^+} (2 \tan^{-1} 1 - 2 \tan^{-1} a)$$

$$+ \lim_{b \rightarrow \infty} (2 \tan^{-1} b - 2 \tan^{-1} 1) = 2\left(\frac{\pi}{4}\right) + 2\left(\frac{\pi}{2}\right) - 2\left(\frac{\pi}{4}\right) = \pi$$

Per-Duet



Thm 6.7 (A Special Type of Improper Integral)

$$\int_1^{\infty} \frac{1}{x^p} dx = \begin{cases} \frac{1}{p-1}, & p > 1 \\ \text{diverges}, & p \leq 1. \end{cases}$$

Note: Thm 6.7 will be used in the Integral Test (積分測試法) for determining the convergence of an infinite series (無窮級數的收斂性), see Section 7.3 for details.



$\text{If: } \textcircled{1} \text{ If } p=1, \text{ then } \int_1^{\infty} \frac{1}{x} dx = \lim_{b \rightarrow \infty} \left(\int_1^b \frac{1}{x} dx \right) = \lim_{b \rightarrow \infty} (\ln|b| - \ln 1)$
 $= \infty. \text{ (diverges.)}$

$\textcircled{2} \text{ If } p \neq 1, \text{ then } \int_1^{\infty} x^{-p} dx = \lim_{b \rightarrow \infty} \left(\frac{1}{-p+1} x^{-p+1} \Big|_1^b \right)$
 $= \lim_{b \rightarrow \infty} \left(\frac{1}{-p+1} b^{-p+1} + \frac{1}{p-1} \right) = \begin{cases} \frac{1}{p-1}, & p > 1 \\ \infty, & p < 1. \end{cases}$

$\text{So, from } \textcircled{1} \text{ and } \textcircled{2} \Rightarrow \int_1^{\infty} \frac{1}{x^p} dx = \begin{cases} \frac{1}{p-1} & \text{if } p > 1 \\ \text{diverges} & \text{if } p \leq 1, \end{cases}$



Comparison Theorem for Improper Integrals

Thm (瑕積分的比較定理)

Suppose that f and g are **conti. on** $[a, \infty)$ with $f(x) \leq g(x) \quad \forall x \in [a, \infty)$.

$$(1) \quad \int_a^{\infty} g(x) \, dx \text{ converges} \implies \int_a^{\infty} f(x) \, dx \text{ converges.}$$

$$(2) \quad \int_a^{\infty} f(x) \, dx \text{ diverges} \implies \int_a^{\infty} g(x) \, dx \text{ diverges.}$$



Example (補充題)

Does the integral $\int_1^{\infty} e^{-x^2} \tan^{-1} x dx$ converge or diverge?

Sol: Since $x^2 \geq x$ for all $x \geq 1$, it follows that $-x^2 \leq -x$ for all $x \geq 1$ and hence we know that

$$f(x) \equiv e^{-x^2} \tan^{-1} x \leq \frac{\pi}{2} e^{-x} \equiv g(x) \quad \forall x \in [1, \infty).$$

In addition, since the improper integral

$$\int_1^{\infty} g(x) dx = \frac{\pi}{2} \int_1^{\infty} e^{-x} dx = \frac{\pi}{2} \lim_{b \rightarrow \infty} (-e^{-x}) \Big|_1^b = \frac{\pi}{2e},$$

it follows from the Comparison Thm that the improper integral

$$\int_1^{\infty} f(x) dx = \int_1^{\infty} e^{-x^2} \tan^{-1} x dx \text{ converges.}$$



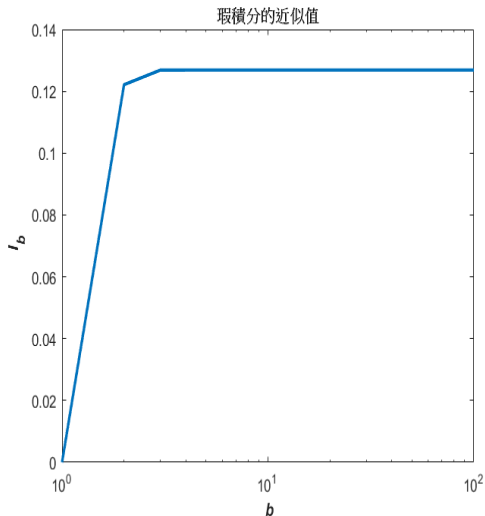
Numerical Results (1/2)

- What is the true value of $I = \int_1^{\infty} e^{-x^2} \tan^{-1} x dx$?
- Although it is usually difficult to get the true value of I , we may apply the technique of numerical integration (數值積分) to compute its approximate value.
- If $I_b \equiv \int_1^b e^{-x^2} \tan^{-1} x dx$ for $b \geq 1$, we see that

$$I = \lim_{b \rightarrow \infty} I_b \approx 0.1268694335685518.$$



Numerical Results (2/2)



Thank you for your attention!

